

Consider the following set of constraints:

$$x_1 + x_2 \geq 1,$$

$$x_1 + x_2 \leq 2,$$

$$x_1 \geq 0,$$

$$x_2 \geq 0.$$

1. Draw the polyhedron in $\mathbb{R}^2$.
2. Write the polyhedron in standard form.
3. Enumerate all potential bases. For each of them, check if it is a valid basis. If so, if it is feasible. If so, identify the corresponding vertex on the polyhedron.